DEEPFAKE INVESTIGATION PORTAL

Abstract

The rapid advancement of artificial intelligence has significantly improved the creation of realistic digital media; however, it has also led to the widespread emergence of deepfakes, which are AI-generated or manipulated images, videos, and audio recordings that closely resemble authentic content. These manipulated media files pose serious threats by enabling misinformation, identity theft, financial fraud, political manipulation, cybercrime, and the spread of fake news. As deepfakes become increasingly difficult to distinguish from genuine content, law enforcement agencies, journalists, cybersecurity professionals, and digital forensic investigators face major challenges in verifying the authenticity of digital evidence. Existing tools often focus on a single media type, provide limited forensic capabilities, or require advanced technical expertise, making the investigation process time-consuming and inefficient. Therefore, there is a need for a centralized and intelligent platform that can accurately detect deepfakes, support forensic analysis, preserve digital evidence, and assist investigators in producing reliable investigation reports.

The proposed Deepfake Investigation Portal is a web-based digital forensic platform developed to detect and analyze AI-generated or AI-manipulated multimedia content, including images, videos, and audio files. The system aims to provide a centralized environment where investigators can securely upload suspicious media, perform automated deepfake detection, and manage the complete investigation process from evidence submission to report generation. It is intended for use by law enforcement agencies, cyber forensic experts, cybersecurity professionals, journalists, and digital investigators who require reliable tools to verify the authenticity of digital media. The portal integrates Artificial Intelligence (AI) and Machine Learning (ML) models to identify AI-generated content, along with Computer Vision techniques for image and video analysis, Audio Forensics for detecting AI-generated or cloned voices, and Metadata Analysis to extract file information and support digital evidence management. Developed as a web-based application, the system also utilizes Cloud Computing for scalable deployment and storage, Docker for containerized application deployment, and cryptographic hashing techniques to maintain the integrity of digital evidence. By combining these technologies, the proposed solution provides an efficient, secure, and reliable platform for investigating AI-generated media and supporting digital forensic investigations.

The development of the Deepfake Investigation Portal is planned in two phases to ensure systematic implementation and future scalability.

Phase 1: The initial phase focuses on developing the core functionalities of the system. This includes implementing user authentication and role-based access control for administrators, investigators, cyber forensic experts, and public users. The portal will support investigation case management, secure uploading of image, video, and audio evidence, evidence storage, metadata extraction, investigation history, activity logging, case tracking, and generation of forensic investigation reports. Basic dashboard interfaces and database management will also be developed to provide a secure and efficient environment for managing digital investigations.

Phase 2: The second phase enhances the system by integrating advanced Artificial Intelligence (AI) and Machine Learning (ML) capabilities for automated deepfake detection. AI models will analyze uploaded images, videos, and audio files to identify AI-generated or manipulated content and generate confidence scores to assist investigators. The system will incorporate computer vision techniques for media analysis, automated report generation, notification services for completed analyses, investigation analytics, and cloud-based deployment using Docker containers to improve scalability and performance. These enhancements will transform the portal into an intelligent digital forensic platform capable of supporting efficient and reliable deepfake investigations.

Outcome

The proposed Deepfake Investigation Portal is expected to significantly improve the process of detecting and investigating AI-generated multimedia by providing a centralized, secure, and intelligent digital forensic platform. By combining artificial intelligence with digital forensic techniques, the system will reduce the time and effort required to analyze suspicious media while improving the accuracy and reliability of investigation results. Features such as automated deepfake detection, secure evidence management, role-based access control, and comprehensive forensic report generation will enable investigators to conduct digital investigations more efficiently and systematically. The portal is expected to support law enforcement agencies, cybersecurity professionals, journalists, and digital forensic experts in combating misinformation, identity fraud, cybercrime, and the misuse of AI-generated content. Ultimately, the system aims to strengthen digital evidence verification, enhance public trust in digital media, and contribute to a safer and more secure digital ecosystem.

Phase 1: Target Users and Functionalities

1. **Administrator**
    - Manage user accounts and roles
    - Approve or deactivate user accounts
    - Configure AI models and system settings
    - Manage datasets
    - Monitor system performance
    - View activity logs
    - Configure security settings
    - Backup and restore database

2. **Investigator**
    - Create investigation cases
    - Upload image, video, and audio evidence
    - Initiate AI-based deepfake analysis
    - View analysis results
    - Add investigation notes
    - Generate and download investigation reports
    - View investigation history

3. **Cyber Forensic Expert**
    - Review AI analysis results
    - Analyze extracted metadata
    - Validate AI-generated findings
    - Add forensic observations
    - Confirm or reject AI detection results
    - Approve final investigation reports

4. **Public User**
    - Register/Login
    - Submit suspicious AI-generated media
    - Upload supporting files
    - Provide incident description
    - Track submission status
    - Receive submission acknowledgment

5. **AI Analysis Engine (System Module)**
    - Validate uploaded media
    - Preprocess image, video, and audio files
    - Detect AI-generated images
    - Detect AI-generated videos
    - Detect AI-generated audio
    - Extract metadata
    - Store analysis results

Phase 2: AI Project Modules

1. **Image Deepfake Detection**
    - Detect AI-generated images
    - Generate confidence score
    - Highlight suspicious regions (if supported by the model)
    - Store detection results in the investigation case

2. **Video Deepfake Detection**
    - Extract video frames
    - Detect manipulated video frames
    - Generate overall confidence score
    - Display video analysis results

3. **Audio Deepfake Detection**
    - Analyze uploaded audio
    - Detect AI-generated speech
    - Generate confidence score
    - Store audio analysis results

4. **AI Report Generation**
    - Summarize AI detection findings
    - Include confidence scores
    - Generate downloadable PDF reports
    - Attach reports to investigation cases

5. **Investigation Analytics**
    - Display analysis statistics
    - Show case status distribution
    - Visualize investigation trends
    - Generate investigation summaries

Technologies Used

1. **Front End**
    - Next.js
    - TypeScript
    - Tailwind CSS
    - HTML5
    - CSS3
    - JavaScript
    - Axios
2. **Back End**
    - Python
    - Django
    - Django REST Framework (DRF)
3. **Database**
    - MySQL
4. **AI / ML**
    - TensorFlow or PyTorch
    - OpenCV
    - Librosa
    - FFmpeg
    - NumPy
    - Pandas
5. **Other Tools**
    - Docker
    - AWS S3
    - Nginx
    - Git & GitHub
    - Postman
    - Visual Studio Code
    - SHA-256 Hashing
    - ReportLab or PDFKit (for PDF generation)